



Public Health Surveillance & Outbreak Tracking

- **The Goal:** Shifting from reactive to proactive healthcare. By analyzing daily admission records, we can identify "spikes" in specific symptoms before they become full-blown epidemics.
- **KPK Context:** Focusing on seasonal endemic patterns like Dengue during the monsoon or Malaria in warmer months.
- **Spatial Tracking:** Identifying clusters in specific union councils or districts to target vaccination and fumigation efforts effectively.



Medical Imaging Pipelines (The DICOM Standard)

- **What is DICOM?:** Digital Imaging and Communications in Medicine. It is the universal format for MRI, CT, and X-ray data.
- **The Architecture:** A DICOM file is more than an image; it contains a "header" with patient metadata (protected by privacy protocols) and a "pixel array" containing high-depth anatomical data.
- **Processing:** Using Python to extract specific "slices," normalize pixel intensities, and prepare the data for advanced segmentation models.



Patient Flow & Hospital Optimization

- The Bottleneck Problem: Large centers like Lady Reading Hospital (LRH) handle thousands of patients daily. Bottlenecks in triage or diagnostic testing can lead to dangerous delays.
- Queueing Theory: Using data to model patient arrival rates (λ) and service rates (μ) to calculate the optimal number of staff required at any given hour
- Outcome: Reducing the "Wait-to-Treatment" ratio to improve patient satisfaction and clinical outcomes.